

titanic

March 11, 2025

1 Reading data

Read the train.csv file as a pandas dataframe.

```
[1]: import pandas as pd

titanic=pd.read_csv('Data/train.csv')
titanic
```

```
[1]:      PassengerId  Survived  Pclass  \
0                1         0        3
1                2         1        1
2                3         1        3
3                4         1        1
4                5         0        3
..          ...     ...     ...
886            887         0        2
887            888         1        1
888            889         0        3
889            890         1        1
890            891         0        3
```

```
      Name      Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris    male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2  Heikkinen, Miss. Laina    female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1
4  Allen, Mr. William Henry    male  35.0    0
..          ...     ...     ...
886  Montvila, Rev. Juozas    male  27.0    0
887  Graham, Miss. Margaret Edith    female  19.0    0
888  Johnston, Miss. Catherine Helen "Carrie"    female   NaN    1
889  Behr, Mr. Karl Howell    male  26.0    0
890  Dooley, Mr. Patrick    male  32.0    0
```

```
      Parch      Ticket    Fare Cabin Embarked
0         0      A/5 21171   7.2500   NaN        S
1         0      PC 17599  71.2833   C85        C
```

2	0	STON/02.	3101282	7.9250	NaN	S
3	0		113803	53.1000	C123	S
4	0		373450	8.0500	NaN	S
..	...		...	...	...	
886	0		211536	13.0000	NaN	S
887	0		112053	30.0000	B42	S
888	2	W./C.	6607	23.4500	NaN	S
889	0		111369	30.0000	C148	C
890	0		370376	7.7500	NaN	Q

[891 rows x 12 columns]

2 Indexing

1. Create a function that returns the name of a passenger given their PassengerId.
2. Create a function that returns the PassengerId of a passenger given their Name.
3. Print a message with the ID of passenger **Montvila, Rev. Juozas** with the following format: 'The ID of passenger Montvila, Rev. Juozas is ##'
4. Print a message with the name of the passenger with ID **42** with the following format: 'The passenger with ID 42 is X'

```
[2]: #1
def id_to_name(df: pd.DataFrame, id: int) -> str:
    """Regresa el nombre del pasajero dado su id

    Args:
        df (pd.DataFrame): El conjunto de datos
        id (int): El id del pasajero

    Returns:
        str: El nombre del pasajero
    """
    return df[df.PassengerId==id].Name.values[0]

#2
def name_to_id(df: pd.DataFrame, name: str) -> int:
    """Regresa el id del pasajero dado su nombre

    Args:
        df (pd.DataFrame): El conjunto de datos
        name (str): El nombre del pasajero

    Returns:
        int: El id del pasajero
    """
    return int(df[df.Name==name].PassengerId.values[0])
```

```
#3
print('The ID of passenger Montvila, Rev. Jouzas is ',name_to_id(titanic,
↪'Montvila, Rev. Juozas'))
print('The passenger with ID 42 is ',id_to_name(titanic, 42))
```

The ID of passenger Montvila, Rev. Jouzas is 887
The passenger with ID 42 is Turpin, Mrs. William John Robert (Dorothy Ann Wonnacott)

5. Print all information about the oldest passenger.

```
[3]: titanic.iloc[titanic.Age.idxmax()]
```

```
[3]: PassengerId      631
Survived              1
Pclass               1
Name      Barkworth, Mr. Algernon Henry Wilson
Sex                male
Age              80.0
SibSp              0
Parch              0
Ticket            27042
Fare              30.0
Cabin            A23
Embarked           S
Name: 630, dtype: object
```

3 Subsetting

We are asked to share data for analysis by a third party. Since our dataset contains personal details, we only want to share with them the following information: ticket classes, fares and port of embarkation. We are asked to deliver a sample of the first 100 rows of this dataset.

6. Create and save the new dataset in **data/port_fares.csv**.

```
[4]: port_fares=titanic.loc[:100,['Pclass','Fare','Embarked']]
port_fares.to_csv('Data/port_fares.csv',index=False)
```

4 Counting

7. We want to know if there were any survivors over the age of 60, print all of their information.
8. How many people over 60 survived?
9. What percentage of people over 60 survived?

5 Women and children first?

10. Find out if women and children were more likely to survive.

11. Write a function that returns the percentage of people that survived from a subset given as a boolean Pandas series.

6 Summarizing

12. What is the median age of the passengers?
13. How many passengers embarked from each port?
14. Generate two hypotheses about how does the survival rate differ among groups of passengers. Write your code to explore both hypotheses.